SUMMARY

- A state of rest or of motion is defined with respect to a frame of reference.
- In Physics, we conventionally follow a right handed Cartesian coordinate system to explain the motion of objects
- To explain linear motion the concept of point mass is used.
- A vector is a quantity which has both magnitude and direction. A scalar has only magnitude.
- The length of a vector is called magnitude or norm of the vector
- In a Cartesian coordinate system the unit vectors are orthogonal to each other.
- Vectors can be added using either the triangular law of addition or the parallelogram law of addition.
- Any vector can be resolved into three components with respect to a Cartesian coordinate system
- The magnitude or norm of a vector $\vec{A}$ is given by $\left|\vec{A}\right| = A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
- If two vectors are equal, then their corresponding individual components should be separately equal.
- The position vector of a particle with respect to a Cartesian coordinate system is given by $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$
- The scalar product of two vectors is defined by $\vec{A}.\vec{B} = AB\cos\theta$ (θ *is the angle between* $\vec{A}$ *and* $\vec{B}$)
- The vector product of two vectors is defined by $\vec{A} \times \vec{B} = \left(AB\sin\theta\right)\hat{n}$. The direction of $\hat{n}$ can be found using right hand thumb rule or right hand cork screw rule.
- In physics, scalar and vector products are used to describe various concepts.
- Distance is the total path length travelled by the particle and displacement is the difference between final and initial positions. Distance is a scalar quantity and displacement is a vector
- Average velocity is defined as $\vec{v}_{avg} = \dfrac{\Delta\vec{r}}{\Delta t}$ and instantaneous velocity is defined as $\vec{v} = \lim\limits_{\Delta t \to 0} \dfrac{\Delta\vec{r}}{\Delta t} = \dfrac{d\vec{r}}{dt}$. Both average velocity and instantaneous velocity are vector quantities.
- Momentum is defined as $\vec{p} = m\vec{v}$.

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